Algorithms and Analysis

Lesson 32: *Quiz*

Course Quiz

Question 1

- What is the time complexity of TSP using exhaustive search?

1. $\Theta(2^n)$
2. $\Theta(n\ \log(n))$
3. $\Theta(n!)$
4. $\Theta(n^2)$

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- Answer: 3. $\Theta(n!)$

Question 2

- If it takes insertion sort 1 minute to sort 100 000 elements approximately how long will it take to sort 1 million elements?

  1. 1.1 seconds
  2. 10 seconds
  3. 100 seconds
  4. 11 seconds

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- Answer: 3. 100 seconds

Question 3

- What is the commonly used instruction for putting things onto and taking the things of the ADT queue?

1. enqueue and dequeue
2. add, remove
3. insert, `get_minimum`
4. push, poll

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- Answer: 1. enqueue and dequeue

Question 4

- Which ADT represents a LIFO memory?

  1. stack
  2. heap
  3. queue
  4. priority queue

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- Answer: 1. stack

Question 5

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  2. malloc
  3. free
  4. RAII

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- Answer: 4. RAII

Question 6

- What is the standard mechanism for implementing a variable length array?

1. Add a pointer at the end of current array to new array
2. Use a stack
3. Create twice the memory and copy existing array
4. Use large enough initial array that it fits all possible data

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- Answer: 3. Create twice the memory and copy existing array

Question 7

- What is the closest bound for the amortorized cost of resizing an array?

1. twice the length of data being added
2. $\Theta(\log(n))$ additional memory
3. 3 times the length of data being added
4. $\Theta(n)$ time complexity

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- Answer: 3. 3 times the length of data being added

Question 8

- What is the cost of inserting one linked list into another?

1. $\Theta(1)$
2. $\Theta(n\ \log(n))$
3. $\Theta(n)$
4. $\Theta(n^2)$

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- Answer: 1. $\Theta(1)$

Question 9

- What is the cost of finding an element in a linked list?

  1. $\Theta(n^2)$
  2. $\Theta(n)$
  3. $\Theta(1)$
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- What is the cost of finding an element in a linked list?

1. $\Theta(n^2)$
2. $\Theta(n)$
3. $\Theta(1)$
4. $\Theta(n\ \log(n))$

- Answer: 2. $\Theta(n)$

Question 10

- What is the time complexity of finding the integer power of a number?

1. $\Theta(\log(n))$
2. $\Theta(1)$
3. $\Theta(n \log(n))$
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- Answer: 1. $\Theta(\log(n))$

Question 11

- What is the time complexity of solving the towers of Hanoi?

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2. $\Theta(2^n)$
3. $\Theta(\mathrm{e}^n)$
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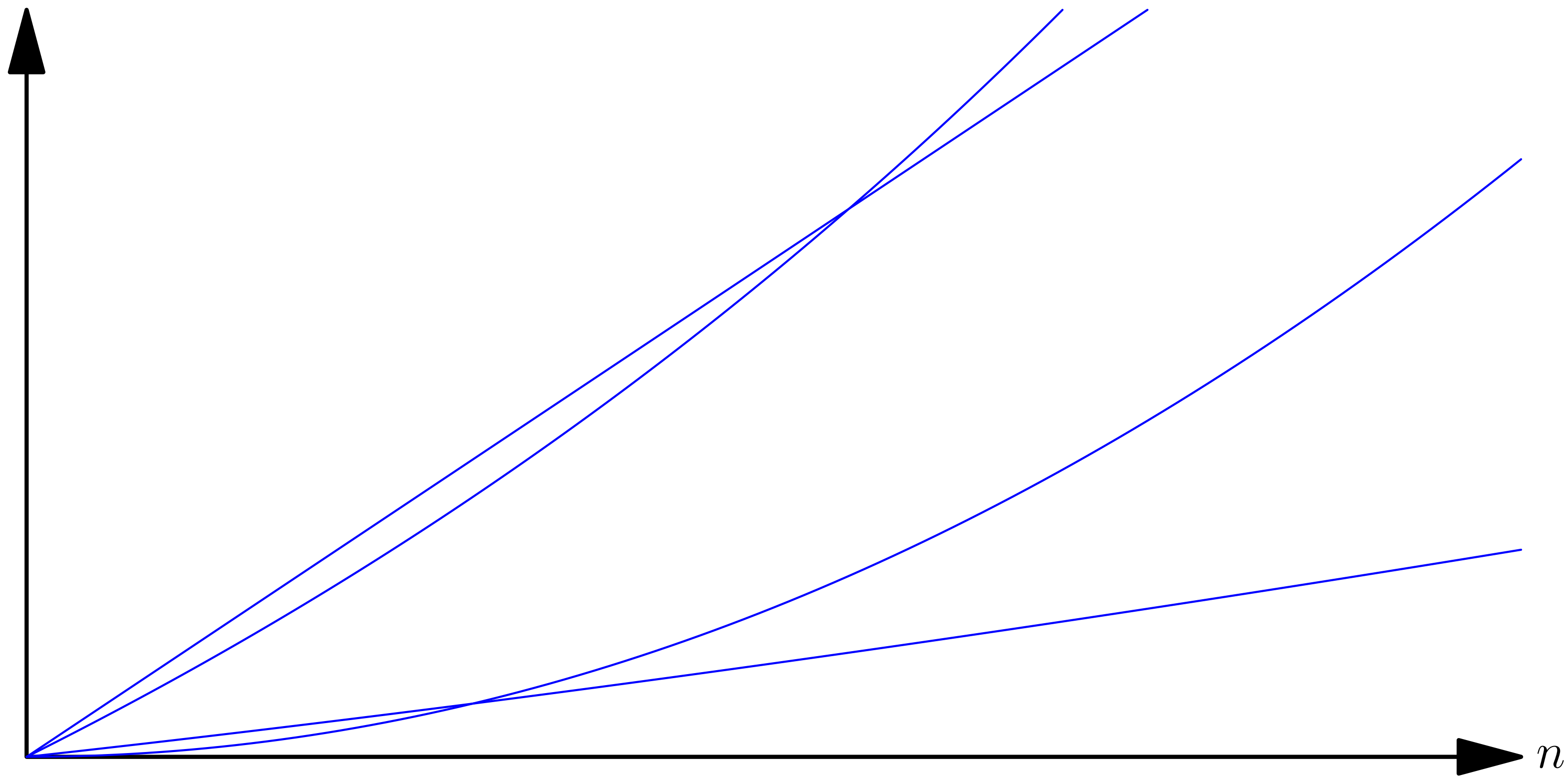
- Answer: 2. $\Theta(2^n)$

Question 12

Which time complexity classes $\Theta(n)$, $\Theta(n\,\log(n))$, $\Theta(n^2)$, $\Theta(\mathrm{e}^{c\,n})$ do the curves belong to?



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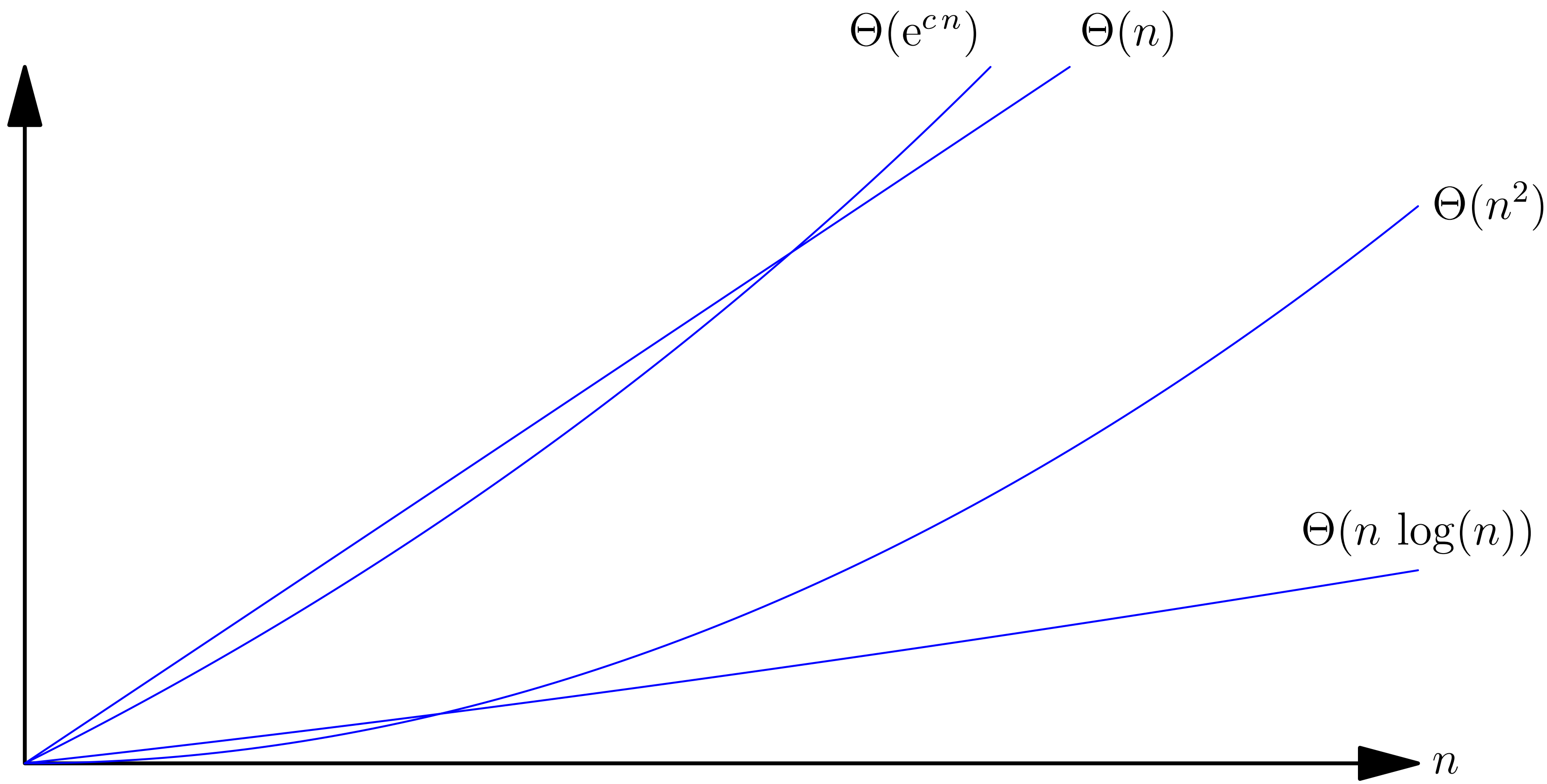
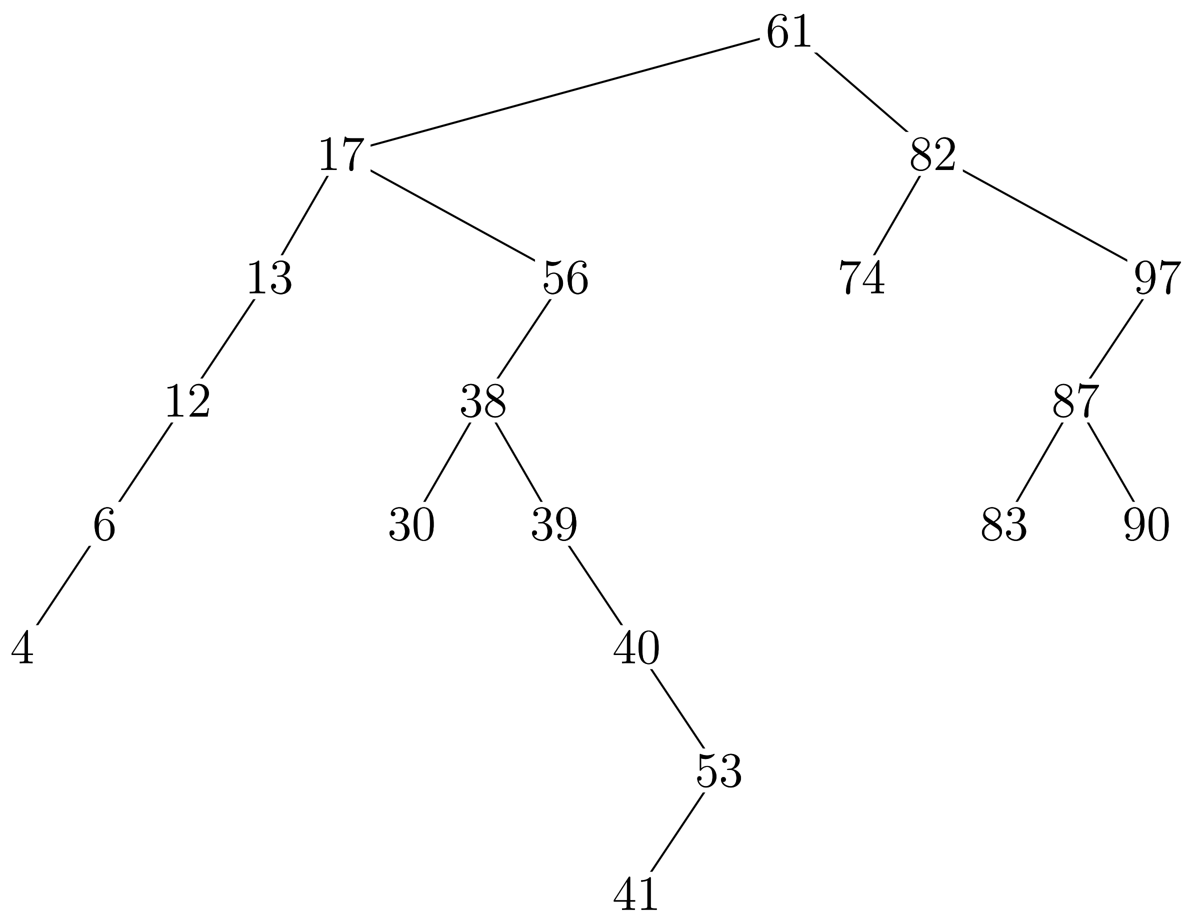


Question 13

Where would 80 be inserted?



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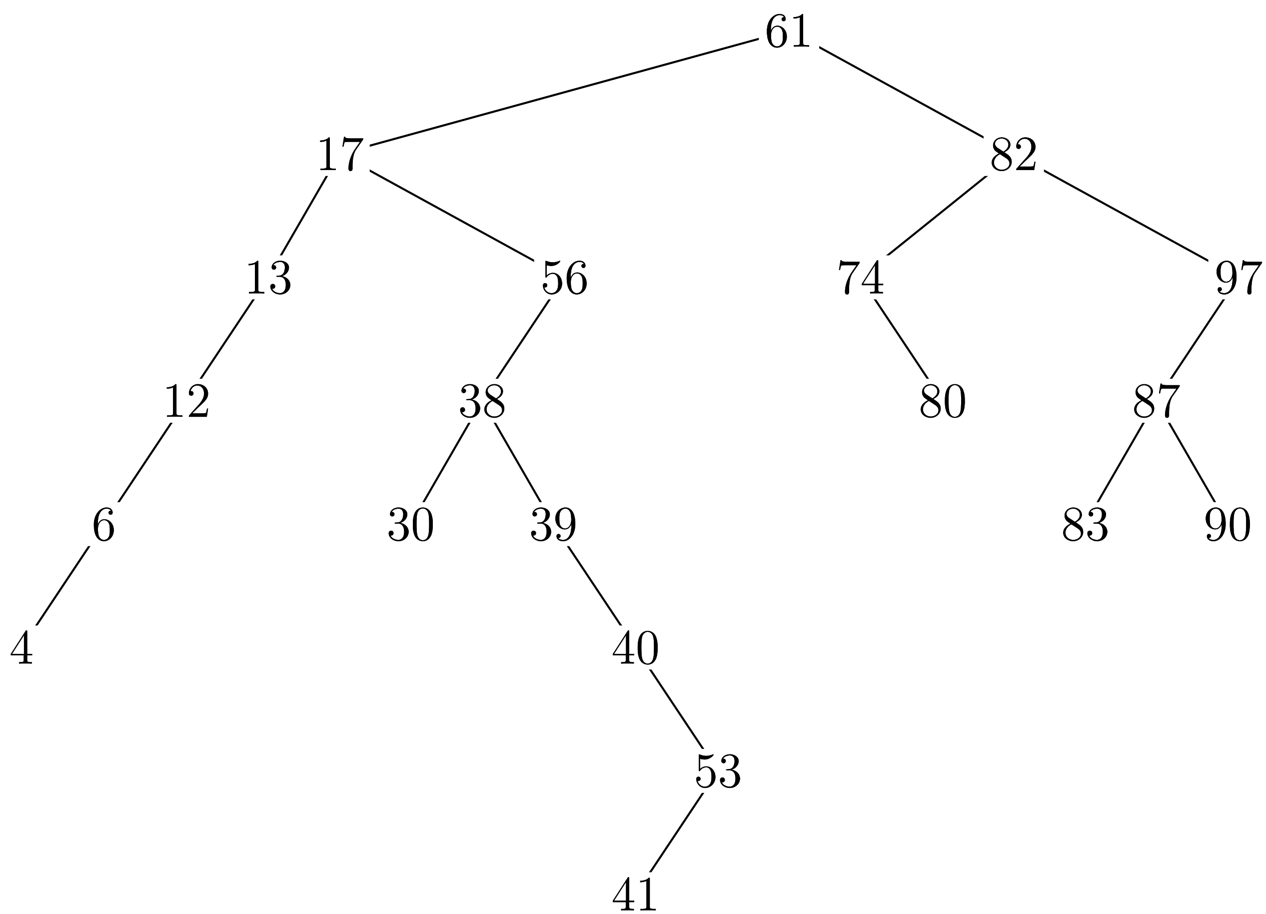

Question 14

- What statement is true of a balanced binary search tree?
  1. insertion is linear time but deletion is constant time
  2. search is constant time
  3. insertion and search are both fast, i.e. $\Theta(\log(n))$
  4. search and insertion depend on the length of the input strings

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- Answer: 3. insertion and search are both fast, i.e. $\Theta(\log(n))$

Question 15

- Which mechanism does C++ use for balancing trees?
  1. Splay Trees
  2. Red-Black Trees
  3. Green-Black Trees
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- Answer: 2. Red-Black Trees

Question 16

- What statement is true about a B-Tree?
  1. B-Trees are balanced trees extensively used in databases
  2. B-Trees are binary trees
  3. B-Trees are part of the standard template library
  4. B-Trees are easy to implement

Question 16

- What statement is true about a B-Tree?

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  2. B-Trees are binary trees
  3. B-Trees are part of the standard template library
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- Answer: 1. B-Trees are balanced trees extensively used in databases

Question 17

- Which statement about **tries** is **false**?
  1. tries are multiway trees
  2. tries are commonly used for spell-checking
  3. tries are known as digital trees
  4. tries use hash tables

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  2. tries are commonly used for spell-checking
  3. tries are known as digital trees
  4. tries use hash tables

- Answer: 4. tries use hash tables is **false**

Question 18

- What statement is true about suffix trees?

  1. A suffix tree is a trie of all suffixes of a text
  2. Suffix trees take $\Theta(n \log(n))$ time
  3. Suffix trees take $\Theta(n \log(n))$ space
  4. Suffix trees are a replacement for B-Trees

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- Answer: 1. A suffix tree is a trie of all suffixes of a text

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- What statement is true of hash-tables?
  1. It is impossible to delete from a hash-table
  2. They have insertion and deletion of $\Theta(1)$
  3. They allow you to iterate over them in sorted order
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- Answer: 2. They have insertion and deletion of $\Theta(1)$

Question 20

- What statement is **not true** of heaps?

  1. Heaps can be viewed as a complete tree
  2. Enqueue and dequeue are both take order $\log(n)$ operations
  3. Heaps are used to implement priority queues
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- Answer: 4. heaps are high performance stack data structures

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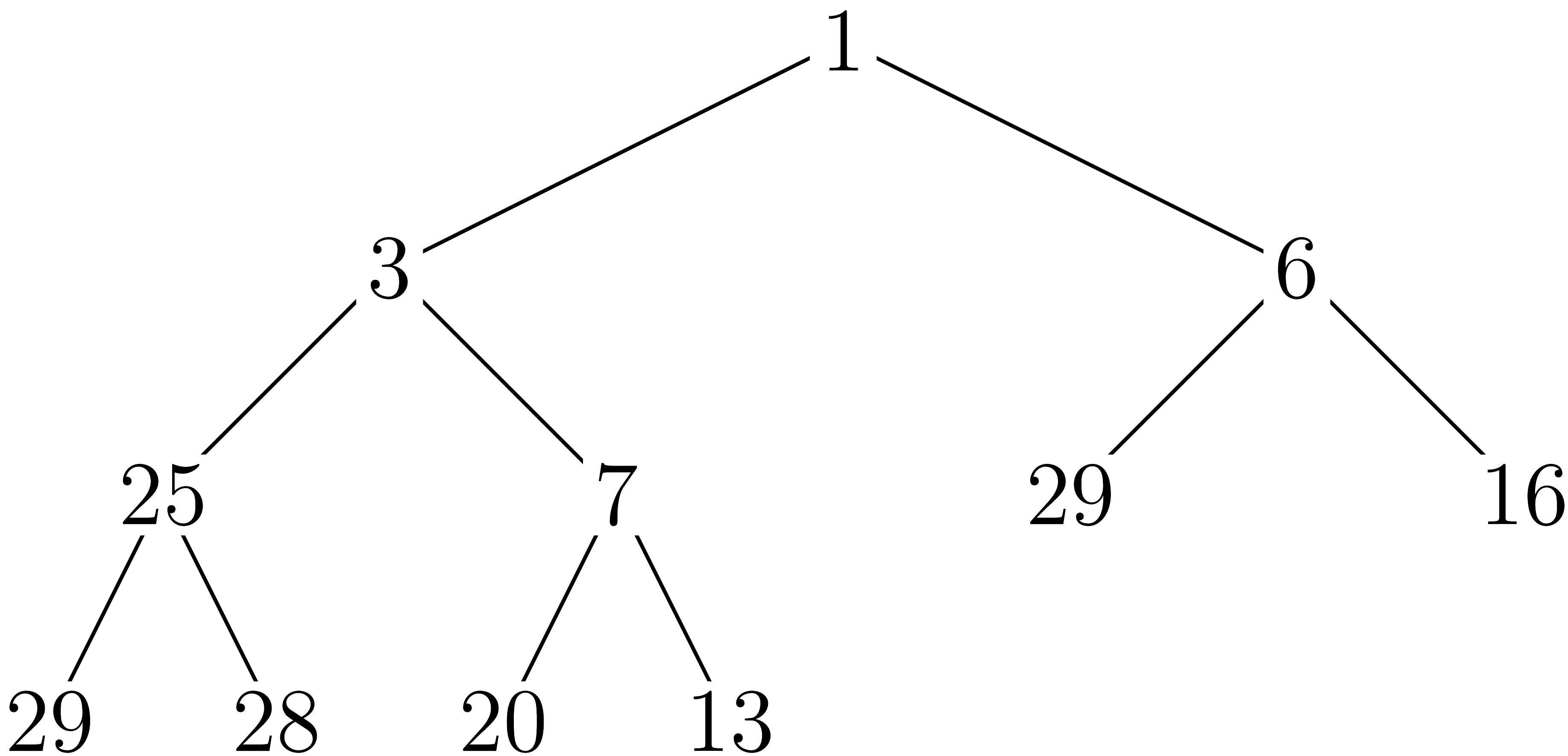
What happens when I add 5 to the heap?



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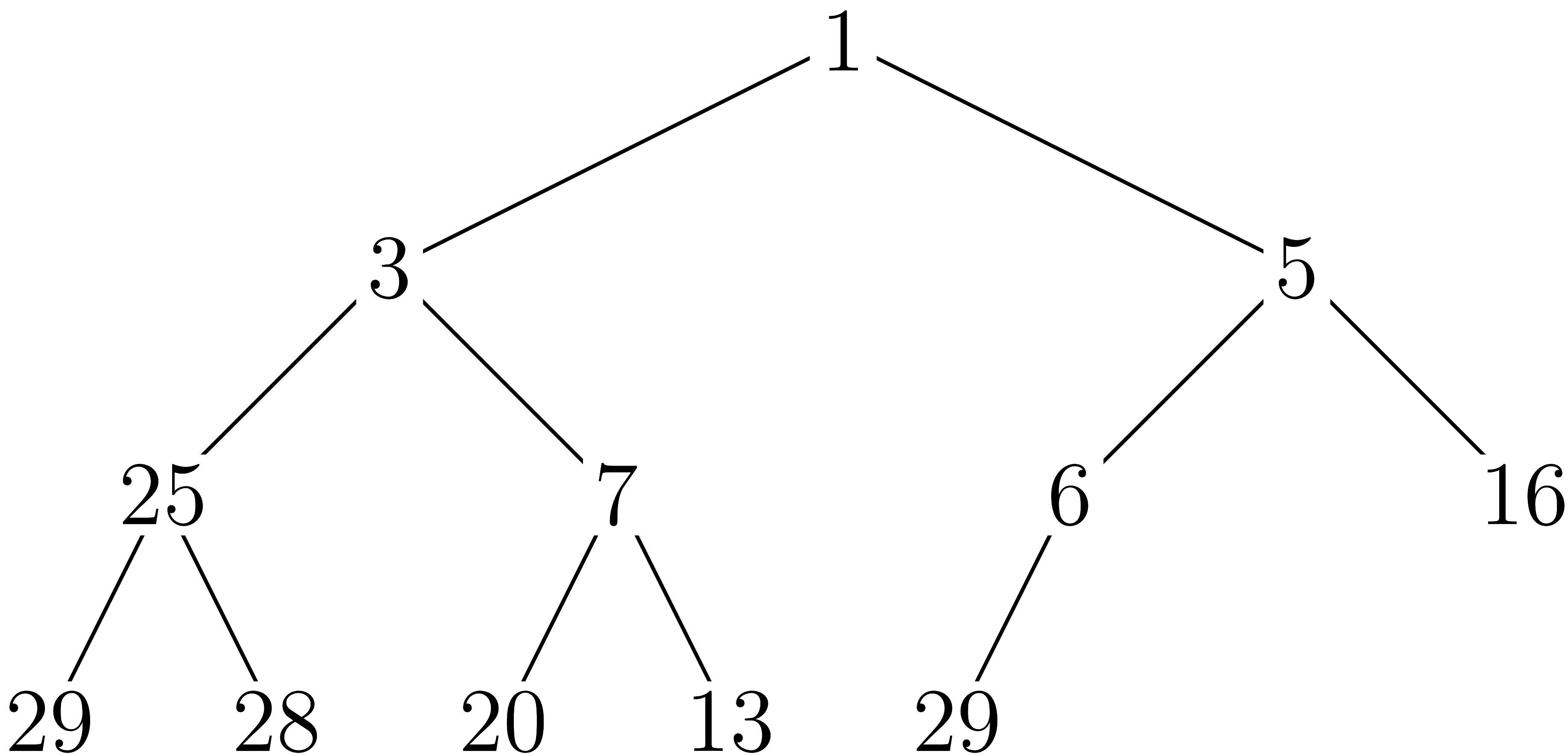
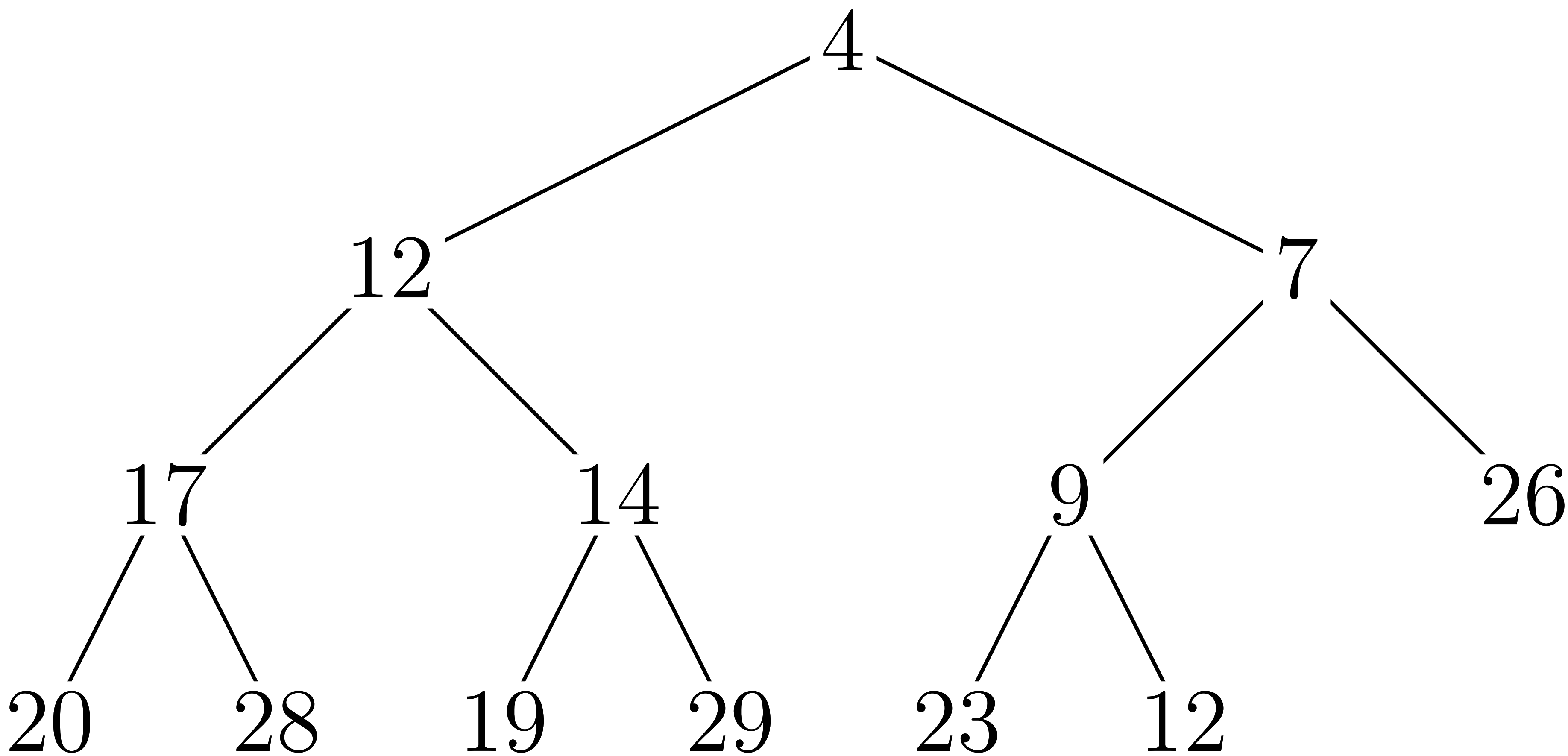


Question 22

What happens when I remove the minimum element from the heap?



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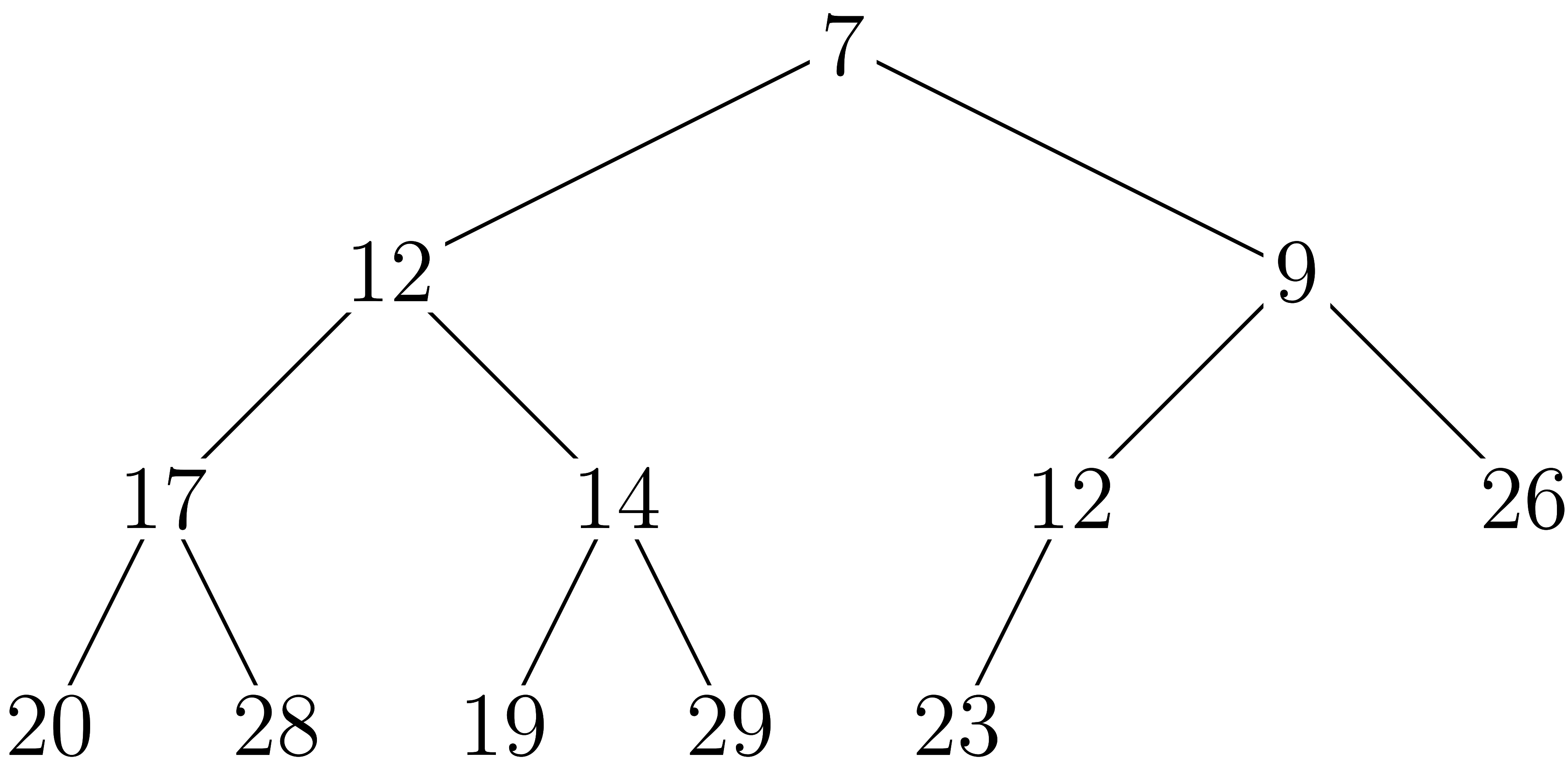

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- Which of the following sorts is faster?
  1. insertion sort
  2. selection sort
  3. bubble sort

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- Answer: 1. insertion sort

Question 24

- Merge sort involves splitting an array in half, merge sorting both halves and then merging the two arrays. Assuming merging two array with combined length n takes $n-1$ operations, what recurrence relation most accurately describes this?

1. $T(n) = 2\,T(n-1)$
2. $T(n) = n\,\log(n)$
3. $T(n) = 2\,T(n/2) + n - 1$
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- Answer: 3. $T(n) = 2\,T(n/2) + n - 1$

Question 25

- What statement is **untrue** about breadth first search?
  1. This can be easily implemented using a queue
  2. This is commonly used for find the shortest route in terms of number of edges
  3. This is the most commonly used algorithm for doing graph traversals
  4. This will search all nodes closest to the start node first

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- Answer: 3. This is the most commonly used algorithm for doing graph traversals is **false**

Question 26

- What best describes branch and bound?
  1. This is used to determine if a graph has articulation points
  2. This is a backtracking algorithm that prunes search depending on whether partial solutions are worse than the current best solution
  3. This makes searching trees polynomial
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- Answer: 2. This is a backtracking algorithm that prunes search depending on whether partial solutions are worse than the current best solution

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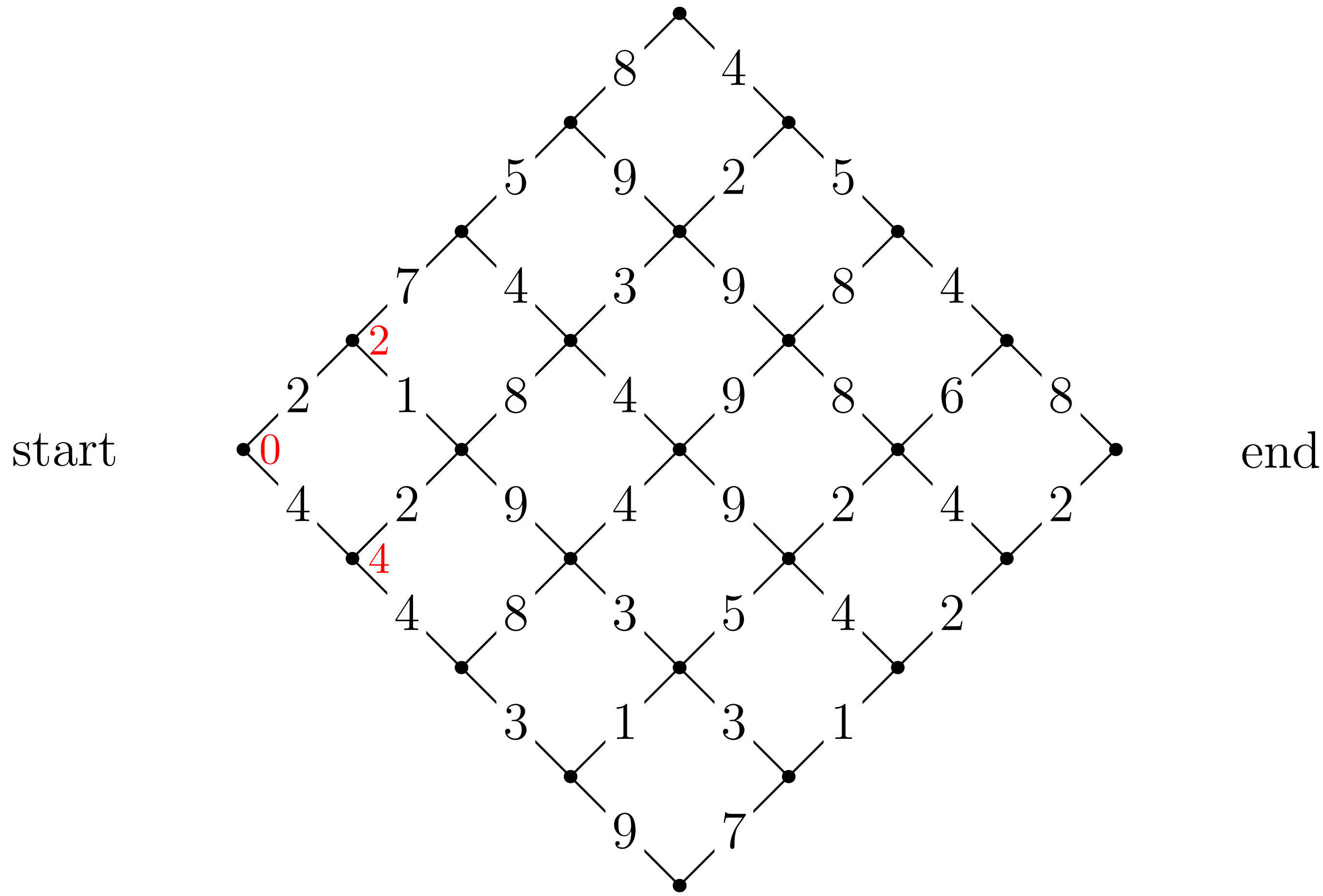
What are the partial costs at the next column?



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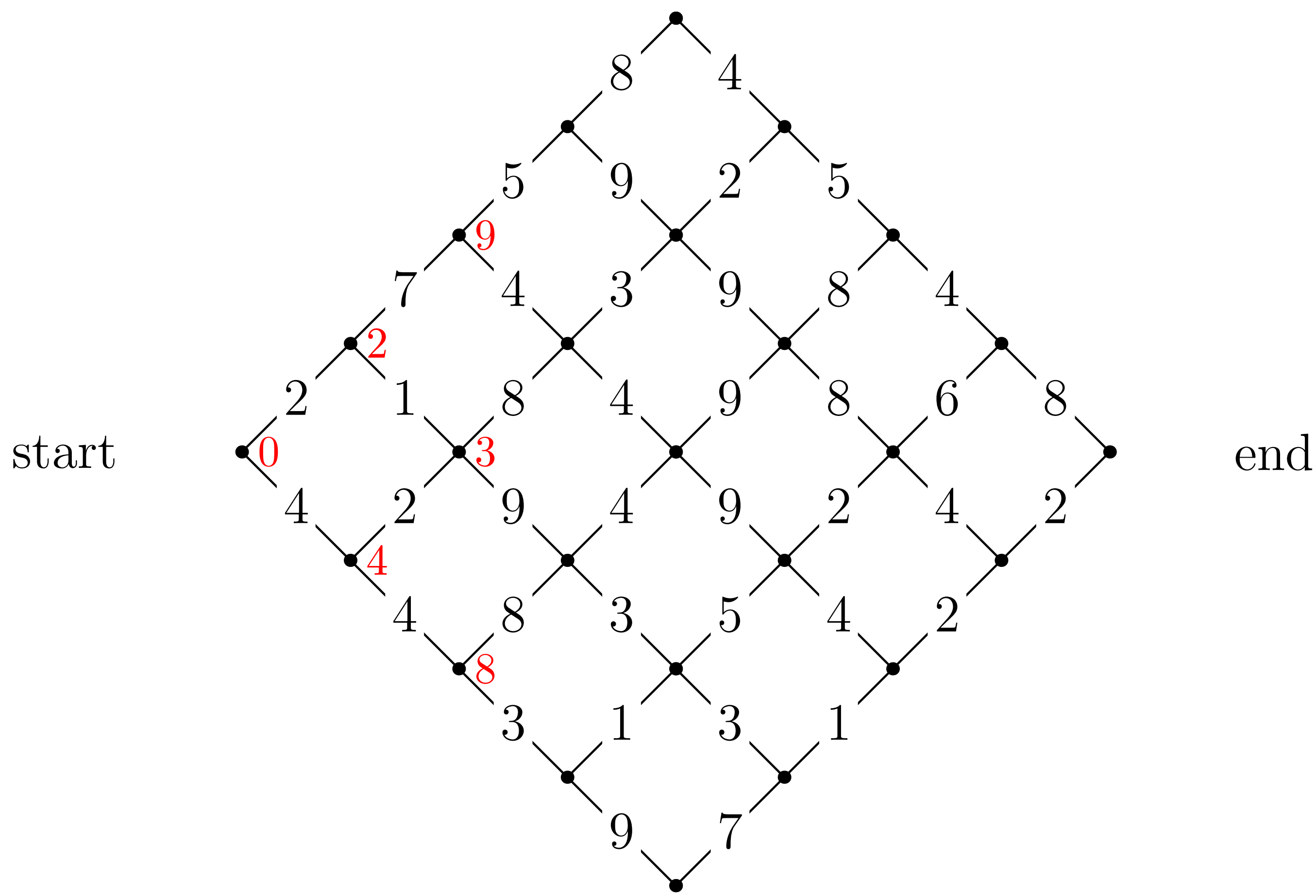

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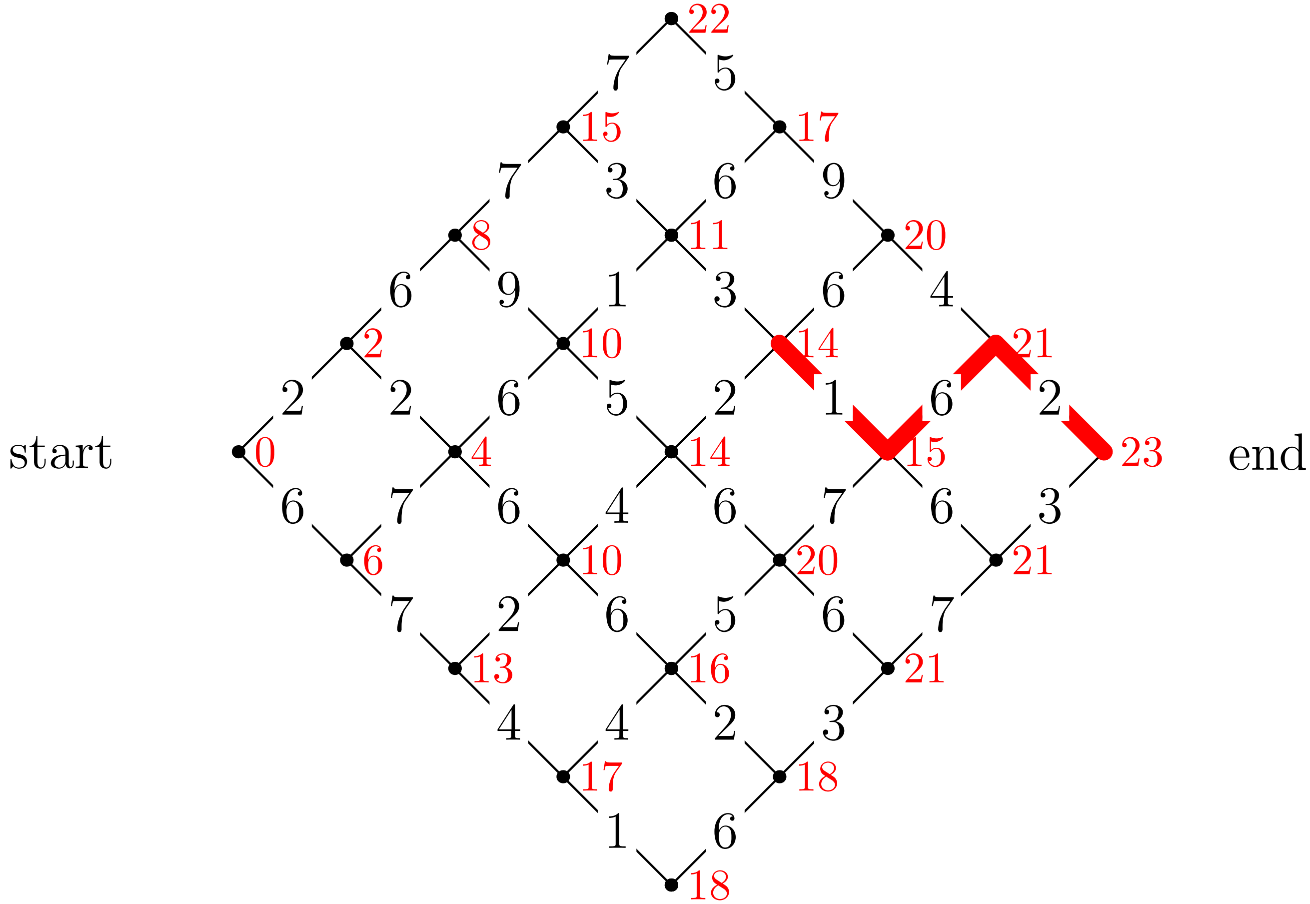
What is the next step in the backwards algorithm?



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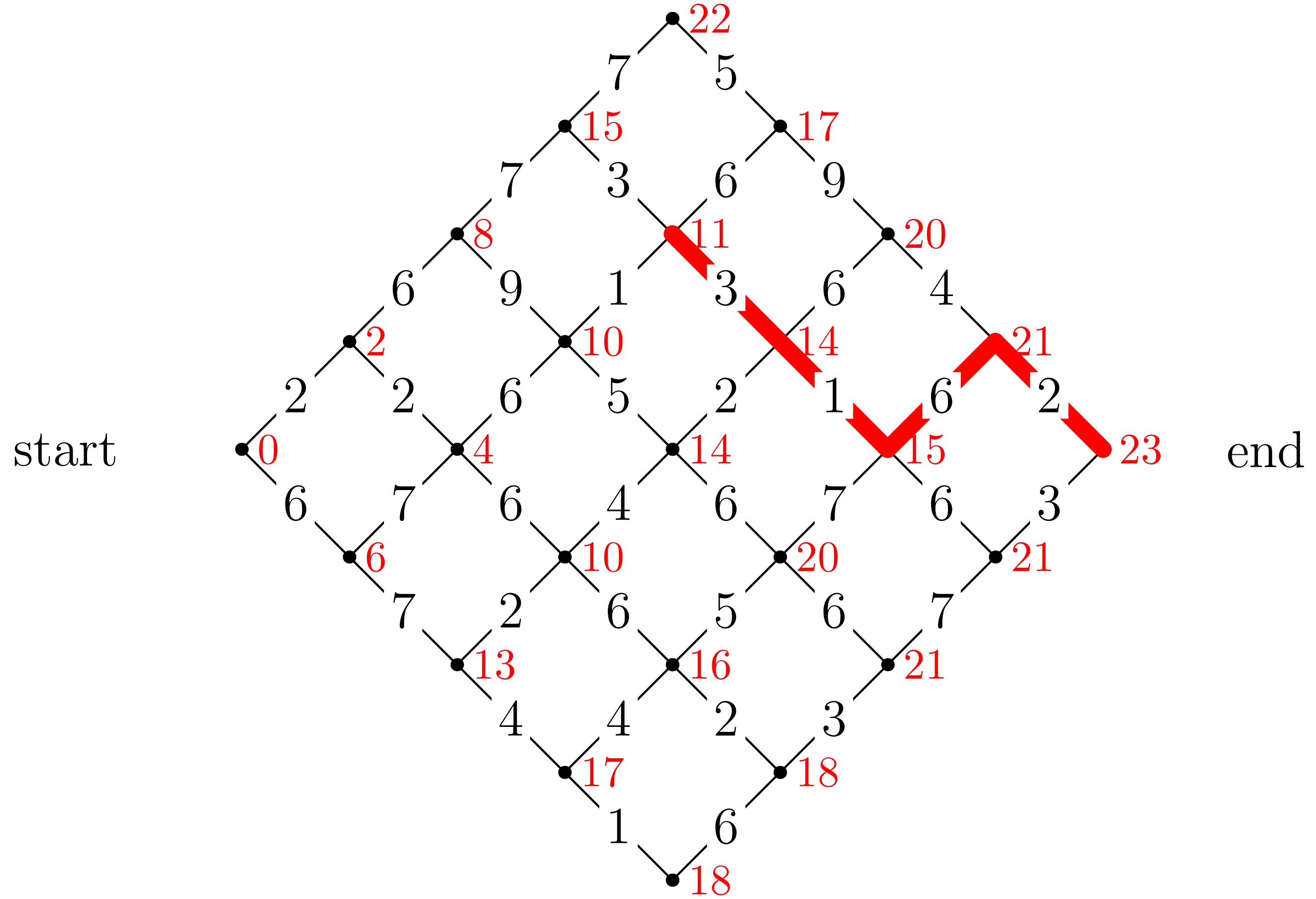

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- What statement is true about wavelet encoding?
  1. They are typically used to do lossless compression
  2. They work by concentrating most of the signal "energy" into a short carrier signal
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- Answer: 2. They work by concentrating most of the signal "energy" into a short carrier signal

Question 30

- What statement is true about problems in NP?

  1. There are no known efficient algorithms to solve any of these problems
  2. They include the problem of whether chess is winnable by white
  3. They are problems that can be done in polynomial time on a non-determinstic Turing machine
  4. They consist of the set of super-polynomial time problems

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- Answer: 3. They are problems that can be done in polynomial time on a non-determinstic Turing machine

Question 31

- What best describes a hill-climbing algorithm?

  1. It is the algorithm used by a bipedal robot to go up hills
  2. It is an algorithm where a step is made in the direction of the gradient
  3. It is a stochastic algorithm which always moves to a better neighbour when one exists
  4. It uses a population of searchers evolving using selection, mutation and crossover

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Question 32

- What statement is true about linear programming?
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  2. It is an NP-hard problem
  3. TSP can be solved using linear programming
  4. The objective function is quadratic, but the constraints are linear

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- Answer: 1. It involves problems with a linear objective function, linear constraints and non-negativity constraints

Question 33

- What does it mean for a linear program to be in normal form?

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2. All inequality constraints except for non-negativity constraints have been turned into equality constraints
3. It does not have any inequality constraints
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- Which algorithm uses a divide and conquer strategy?
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  2. Dijkstra's algorithm
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- Answer: 4. Merge sort

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- Answer: 4. Kruskal's algorithm

Question 36

- In numerical algorithms what causes a large loss in accuracy?

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  2. Adding two numbers with very similar values
  3. Computing the exponential of a number
  4. Subtracting two numbers with very different values

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  4. Subtracting two numbers with very different values
- Answer: 1. Subtracting two numbers with very similar values

Question 37

- What statement is true?

1. If algorithm A is $\Theta(n^3)$ and algorithm B is $\Theta(n^4)$ then algorithm A will run faster than algorithm B
2. If algorithm A and B are both $\Theta(n^2)$ then they take the same time
3. If algorithm A is $O(n^3)$ and algorithm B is $O(n^2)$ then for sufficiently large n, algorithm B will be faster than algorithm A
4. If algorithm A is $\omega(n\,\log(n))$ and algorithm B is $O(n\,\log(n))$ then for sufficiently large n, algorithm B is faster than algorithm A

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- Answer: 4. If algorithm A is $\omega(n \log(n))$ and algorithm B is $O(n \log(n))$ then for sufficiently large n, algorithm B is faster than algorithm A